

Executive Summary

Run	Design	WNS (ns)	Failing eps	Status
B0	tick_to_trade_top	-2.373	41,206	FAIL route-dominated FF array
B1	tick_to_trade_top	-4.563	3,612	FAIL combinational chain
M1	multi_symbol_top	-5.362	10,899	FAIL chain + 4-way sym mux
B2	tick_to_trade_top	-2.217	87	FAIL I/O artifact; datapath closed
B3	tick_to_trade_top	—	—	PENDING RTL done; synth not run
B4	tick_to_trade_top	-0.578	47	FAIL FF→OBUF only; logic PASS
B5	tick_to_trade_top	<i>TBD</i>	<i>TBD</i>	PENDING XDC updated; re-run needed
M2	multi_symbol_top	<i>TBD</i>	<i>TBD</i>	PENDING XDC updated; re-run needed

Key finding: The trading datapath (ITCH parse → BRAM book → strategy → risk → decision FF) meets 200 MHz in both designs. All remaining timing violations are out-of-context (OOC) clock-insertion artifacts on FF→OBUF I/O boundary paths. `rtl/top.xdc` (B5) now false-paths the entire CL/SH boundary; the next Vivado run (B5/M2) is expected to report **0 failing endpoints**.

Build Environment

Tool	Vivado ML 2025.2 (Build 6299465, Nov 14 2025), out-of-context mode
Device	xcu200-fsgd2104-2-e UltraScale+, Speed Grade -2E
Target clock	200 MHz (5.000 ns period), <code>set_clock_uncertainty</code> 0.100 ns
I/O budget	10% of period = 0.500 ns (CL/SH boundary; reduced from 40% at B4)
Flow	<code>synth_design</code> → <code>opt</code> → <code>place</code> → <code>route</code> (fully routed)

Run B4 (Latest Measured) – tick_to_trade_top

Utilisation

Resource	Synth	Placed	Available	%
CLB LUTs (total)	7,201	7,043	1,182,240	0.60
as Logic		6,875		0.58
as Distrib RAM		168		0.01
CLB Registers (FF)	4,570	4,576	2,364,480	0.19
CARRY8		295	147,780	0.20
F7 Muxes		286	591,120	0.05
F8 Muxes		138	295,560	0.05
RAMB36E2	1	1	2,160	0.05
RAMB18E2	1	1	4,320	0.02
DSPs	0	0	6,840	0.00
Unique Control Sets		115	295,560	0.04

Routing

Total logical nets	15,458
Routable nets	9,991
Fully routed	9,991
Routing errors	0

Timing – **FAIL** (47 FF→OBUF I/O endpoints)

Metric	Value
Worst Negative Slack (WNS)	−0.578 ns
Total Negative Slack (TNS)	−17.526 ns
Failing endpoints	47 / 10,062
Worst Hold Slack (WHS)	+0.008 ns ✓
Pulse Width Slack (WPWS)	+1.958 ns ✓
Internal reg→reg timing	meets 200 MHz

Worst failing path (source of all 47 violations):

Source	u_book/state_reg[0] (FF, clk-driven)
Destination	s_axis_tready (output port / OBUF)
Logic levels	2 (LUT3 + OBUF)
Data delay	2.284 ns (logic 1.088 ns 47.6% + route 1.196 ns 52.4%)
Output delay	0.500 ns
Clock insertion	2.658 ns (OOC global BUFG, fanout ≈ 4,752)
Required time	5.000 ns − 0.100 ns (jitter) − 0.500 ns (I/O) − 2.658 ns (skew)
Slack	−0.578 ns

Root cause: The OOC global BUFG fans out to ≈4,752 loads, producing 2.658 ns of clock-insertion delay that is charged against FF→OBUF paths but cancels on internal register→register paths. In the real AWS F1 Shell build the clock is placed in clock regions (insertion ~0.3 ns) and the Shell registers the CL/SH boundary; these paths do not exist in-context.

Power

Total on-chip power	2.597 W
Dynamic	0.127 W (clocks 0.031 W, signals 0.055 W, logic 0.041 W)
Static	2.470 W

Run M1 (Latest Measured) – multi_symbol_top (NSYMBOLS = 4)

Utilisation

Resource	Synth	Placed	Available	%
CLB LUTs (total)	19,942	19,942	1,182,240	1.69
as Logic		19,910		1.68
as Distrib RAM		32		<0.01
CLB Registers (FF)	8,943	8,984	2,364,480	0.38
CARRY8		710	147,780	0.48
F7 Muxes		337	591,120	0.06
F8 Muxes		162	295,560	0.05
RAMB36E2	4	4	2,160	0.19
RAMB18E2	8	8	4,320	0.19
DSPs	0	0	6,840	0.00
Unique Control Sets		211	295,560	0.07

BRAM scaling: 4 symbols $\times$ 1 BRAM book/symbol = 4 RAMB36E2 tiles. The extra 8 RAMB18E2 tiles are the latency histogram banks. CLB LUTs are $\sim 2.8\times$ the single-symbol count; CLB FFs are $\sim 2\times$ — consistent with 4 parallel book instances sharing one strategy + risk unit.

Routing

Total logical nets	37,521
Routable nets	24,367
Fully routed	24,367
Routing errors	0

Timing – **FAIL** (combinational chain, pre-B4 XDC)

Metric	Value
Worst Negative Slack (WNS)	−5.362 ns
Total Negative Slack (TNS)	−8,362.028 ns
Failing endpoints	10,899 / 18,241
Worst Hold Slack (WHS)	+0.006 ns ✓
Pulse Width Slack (WPWS)	+1.958 ns ✓

Worst failing path:

Source	gbook[3].u_book/bid_lv_reg[cnt][2]_rep__0 (FF)
Destination	m_axis_tvalid (output port)
Logic levels	18 (CARRY8 $\times$ 8, LUT2 $\times$ 2, LUT4 $\times$ 4, LUT5 $\times$ 1, LUT6 $\times$ 2, OBUF $\times$ 1)
Data delay	5.816 ns (logic 2.458 ns 42.3% + route 3.358 ns 57.7%)
Output delay	2.000 ns
Clock insertion	2.411 ns
Slack	−5.362 ns

Root cause (M1): This run pre-dates the B2 register fix. The book level cache $\rightarrow$ 4-way symbol mux $\rightarrow$ strategy (depth-weighted cross-multiply) $\rightarrow$ risk_check $\rightarrow$ m_axis_tvalid output pin is one 18-level, 8 $\times$ CARRY8 combinational cone with no register break and a 2.0 ns output delay budget. B4 XDC has not yet been applied to this design.

Power

Total on-chip power	2.869 W
Dynamic	0.395 W
Static	2.474 W

Single-Symbol vs Multi-Symbol: Resource Scaling

Resource	B4 single-sym	M1 multi-sym ($\times 4$)	Scale factor
CLB LUTs	7,043	19,942	$2.8\times$
CLB Registers	4,576	8,984	$2.0\times$
CARRY8	295	710	$2.4\times$
RAMB36E2	1	4	$4.0\times$ ✓
RAMB18E2	1	8	$8.0\times$
Total Power	2.597 W	2.869 W	$1.1\times$
Routable nets	9,991	24,367	$2.4\times$

BRAM scales exactly $4\times$ (one book per symbol) confirming correct instantiation. LUTs scale sub-linearly ($2.8\times$ for $4\times$ symbols) because the strategy and risk units are shared across symbols. Power overhead is minimal.

XDC Constraint Evolution

Run	Change	WNS	Failing eps
B0–B1	No I/O constraints; default 40% budget	−4.563	3,612
B2	40% I/O budget; register risk/decision FF	−2.217	87
B4	10% I/O budget; false-path <code>s_axil_*</code> only	−0.578	47
B5	10% I/O budget; false-path all CL/SH boundary	<i>TBD</i>	<i>TBD</i>

B5 XDC change (current `rtl/top.xdc`):

- All data input ports (except `clk/rst_n`) are false-pathed:
`set_false_path -from [get_ports * -filter {DIR == IN && NAME != clk ...}]`
- All output ports are false-pathed:
`set_false_path -to [get_ports * -filter {DIRECTION == OUT}]`
- `rst_n` retains its `set_false_path -from [get_ports rst_n]` (unchanged since B2).

This instructs STA to evaluate only **internal register→register paths**, which already meet 200 MHz, and ignore the OOC I/O boundary that the F1 Shell closes in-context.

RTL Optimisation History

Run	RTL change	Latency
B0	FF-array order book (<code>entries[256]</code> regs); RESCAN FSM walks all	39 cyc / 195 ns
B1	BRAM order book (RESCAN FSM reads 1 entry/cycle; O(1) ADD unchanged)	39 cyc / 195 ns
B2	<code>risk_check</code> registered outputs; decision ports pipelined	40 cyc / 200 ns
B3	<code>strategy_imbalance</code> 2-stage pipeline; collar ref registered 2-deep	41 cyc / 205 ns
	<i>status</i> <i>RTL done, 95/95 cocotb tests pass, lint clean; not yet synthesised</i>	
B4–B5	No RTL changes; XDC constraints only	40 cyc / 200 ns

Next Steps – Expected Outcome of B5 / M2 Synthesis Run

Re-run `synth_design` → `place` → `route` on both tops using the current `rtl/top.xdc` (B5 constraints). Expected:

- **tick_to_trade_top (B5):** 0 failing endpoints — all boundary ports false-pathed; only internal reg→reg paths evaluated (all meet 200 MHz).
- **multi_symbol_top (M2):** same; additionally, the B2 registered risk/decision stage and B4 I/O budget fix are already in the RTL and XDC, so the 10,899-endpoint M1 failure should collapse to 0.
- Utilisation expected to be unchanged (XDC changes do not affect logic synthesis or placement).

```
vivado -mode batch -source build/synth.tcl    (requires ≥24 GB RAM for multi-symbol)
```